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PROJECT AND TEAM INFORMATION

Project Title

ExamLens: A Syllabus and Previous-Year Question Paper Analyzer

Team Name: *Team #*	*FRACTAL VERSE(DSCPP-2026-T309)*
Team member 1 (Team Lead)	*NAURIYAL, MADHAV – 2510011324* *madhavnauriyal2401@gmail.com* 
Team member 2	*ISHANT–2510370272* *ishantbahukhandi.8@gmail.com* 
Team member 3	*SHARMA,AARAV—251001273* *Aaravsharma2642@gmail.com* 
Team member 4	*SINHA,SASMIT–2510012002* *sasmitsinha2009@outlook.com* 

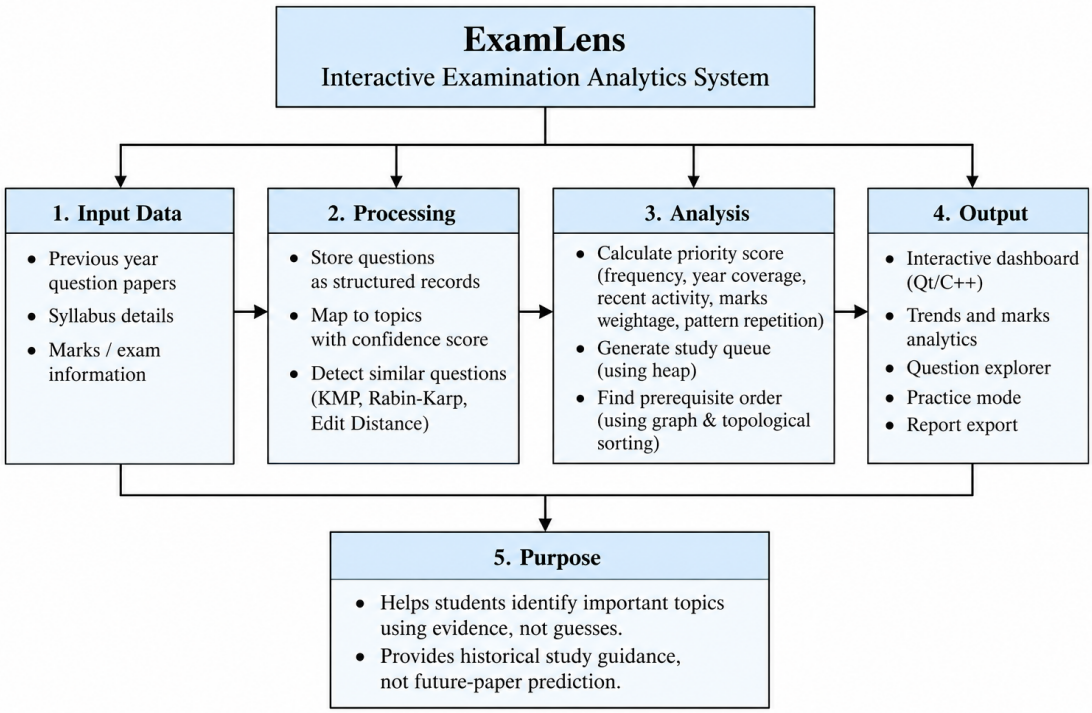
PHASE-II PROJECT PROGRESS DESCRIPTION (35 pts)

Project Abstract

ExamLens is an interactive examination analytics system that helps undergraduate students replace manual counting of previous year questions with objective, evidence-based study priorities. Upgraded from the Phase-I command-line tool, it organizes the syllabus as a Subject, Unit, Topic and Subtopic hierarchy and stores every question as a structured record. Questions are mapped to topics with confidence scores, and low-confidence cases go to manual review. The system detects exact, normalized and similar question patterns using KMP, Rabin-Karp and Edit Distance, supported by a hash table, Trie and ordered structures for fast indexing and search. An explainable priority score combines frequency, year coverage, recent activity, marks weightage and pattern repetition, while a heap generates the study queue. A graph with topological sorting suggests prerequisite-based study order. Results appear in an interactive Qt/C++ dashboard with trends, marks analytics, a question explorer, practice mode and report export. ExamLens offers historical study guidance, not future-paper prediction.

Updated Project Approach and Architecture

the approach was to build a strong, reusable data structures core first, and then use it to power a complete examination analytics application. The system has four layers. The data layer loads the syllabus and previous year questions from text/CSV files into a Subject → Unit → Topic → Subtopic → Question hierarchy and validates them for missing marks, years and duplicates. The core engine in C/C++ uses hash table for searching, a Trie with KMP and Rabin-Karp for mapping questions to topics with confidence scores, and a max heap for the Top-K study queue. The analysis layer in C++ OOP uses Parser, Analyzer and ReportWriter classes to detect exact, normalized and similar repeated questions, calculate an explainable priority score, and use a graph with topological sort for study order. A Qt/C++ dashboard presents the results. Since ExamLens is a standalone, offline, single-user program with all data in memory, no network communication protocol is involved. The libraries used are the C++ standard library (iostream, fstream, sstream, vector, string, unordered map, algorithm) and Qt.



Tasks Completed

Task Completed	Team Member
1:Built the PDF-to-text pipeline with normalization for clean syllabus and question data	Aarav
2:Implemented the hash map, Trie and KMP matching in C	Madhav
3:Built question-to-topic mapping with confidence scores and a review queue	Ishant
4:Added data validation for missing marks, missing years and duplicates	Aarav
5:Built repetition detection and pattern grouping	Madhav
6:Built the analytics engine, priority scoring and heap-based study queue	Sasmit
7:Built the prerequisite graph with BFS, DFS and topological sort	Ishant
8:Built the Qt dashboard and integrated all modules,OPPS concepts in the program	Sasmit, Aarav

Challenges/Roadblocks

-Messy PDF text: Extracted text had broken lines, headers and odd symbols. We cleaned and normalised it, then fixed the rest by hand.
-Unclear topic matches: Some questions match no topic or several. Each mapping gets a confidence score, and weak ones go to manual review.
-Differently worded repeats: Exact matching misses similar questions. We detect repeats at three levels using Rabin-Karp and Edit Distance.
-Benchmarking: KMP may show little gain on short text. We plan to time the algorithms on growing datasets and report the real results.

Tasks Pending

Task Pending	Team Member (to complete the task)
-Interactive UI: Add better filters, clickable charts and smoother navigation across screens.	-SASMIT
-Frequency accuracy: Fix wrong mappings to reduce the gap between program counts and manual counts.	-AARAV
-KMP vs Rabin-Karp: Time both on growing datasets and compare the results with a complexity table.	-ISHANT
-Verification: Run the full pipeline on a real PYQ dataset and check the output against manual counts.	-MADHAV
-Merge to main: Merge develop into main after the integrated test and demo pass.	-MADHAV AND ISHANT

Project Outcome/Deliverables

The main goal of ExamLens is to make tool for the study better by turning their syllabus and past question papers into a list of important topics, ranked from most to least important. It does this automatically, so students don't have to count things by hand. This project includes methods to convert PDFs to text, organize the data using special computational structures called hash maps, tries, and KMPs, and match questions and topics using confidence scores. It also has a system to check for errors such as missing grades, missing years, and duplicate questions. Using techniques called BFS, DFS, and topological sorting, this tool can find repeating patterns, analyze the most important topics, and organize the order of your research. All parts are connected to an easy-to-use dashboard on your computer screen. The package includes code, sample data, and instructions. This projects help what they need to study based on past exams, but does not try to guess future questions.

Progress Overview

About three-fourths of the project is finished. All eight modules are made and connected: the data flow, main C code, topic matching, checking for errors, finding repeated items, scoring and study list, prerequisite map, and the Qt dashboard. The project is doing better than planned on the map part and the dashboard, which were supposed to be finished later. The only part that is taking longer is the accuracy: the topic counts still have some differences from the manual counts because some keywords can be confusing. We are fixing this by adding shortcut names and letting one keyword match multiple topics. The remaining work includes testing, practice mode, final checks, fixing details, and writing instructions, which will be done in the next phase.

Codebase Information

https://github.com/Sasmit33/ExamLens-Syllabus-PYQ-Analysis-System

Testing and Validation Status

Test Type	Status (Pass/Fail)	Notes
Data loading and validation	Pass	The papers are loading just fine
main structures	Pass	Research, keyword matching, and working with Top-K.
Figuring out how topics relate to each other with review	Pass	Variants map to one topic, unmatched go to queue.
comparison of result	Fail	Some words match more than one topic, so a few results are wrong
Connecting the user-facing	Pass	Dashboard shows real output.
Graphs,dashboard	Pass	All modules run from file load to dashboard display.

Deliverables Progress

The dataset is prepared, comprising the syllabus along with at least five cleaned exam papers from previous years. The core components of the program that manage key tools such as hash maps, Trie, heap, sorting, and KMP have been completed.These components were tested for errors, and faulty data rows are flagged rather than deleted. Linking subjects to the review list, detecting duplicates (both exact and near matches), and assigning importance-based scores using the Top-K list and Additionally, the creation of reports (report.md and totals.csv) is still in progress, since they must align topic counts with manual counts and fine-tune the scoring. The user interface built with Qt is also under development, featuring three screens.(Load, Ranked topics, and Topic details) are linked to the real backend. Instructions and explanations are also being written. Certain features are scheduled for the next stage, Phase III, including a unique method of studying with graphs, practice mode, charts, a question explorer, and tools for checking similarity and comparing speed.